System Subsystem Specification

for the

The Balance Project

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Prepared by:

Vinay Agarwal
Balance Project

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POC: Undefined Gpl, undefined.l.gpl.civ@us.navy.mil

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DOCUMENT CHANGE HISTORY

The following table is a simple list of released revisions sent for review. Records of reviews and the review artifacts are saved with reviewer information in the The Balance Projectartifact repository.

Change Record

Date	Version	Author(s)	Change Reference
13 Sep 2025	P1	Vinay Agarwal	Preliminary DRAFT version
5 Dec 2025	P1	Vinay Agarwal	Final Draft of the system

Draft P1 Preliminary version of this document.

1. Baseline Document draft 1
2. Final fix and update from the first draft

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

CHAPTER 1

Scope

This document provides the System / Subsystem Specification (**SSS**) for the Balance System. The system will be referred to as the Balance System.

Acronym, used for testing, is **.PDF**. Acronym, used for testing, is **BioAPI**. Acronym, used for testing, is **MISO**. Acronym, used for testing, is **UAV**. Acronym, used for testing, is **TOC-L**. Acronym, used for testing, is **USN**. Acronym, used for testing, is **USAF**. Acronym, used for testing, is **USSF**. Acronym, used for testing, is **USMC**. Acronym, used for testing, is **USArmy**.

Acronym, used for testing, is **Garmin**. Acronym, used for testing, is **JIDA**.

1.1 Identification

The Balance System described in this document shall be known as Balance System version 1. However, the **SSS** described herein shall be applicable to pre-releases such as Beta-releases for a phased release as listed for each requirement. The major system interfaces and capabilities are fully specified in Chapter 3.

1.2 System Overview

Figure 1 shows the high-level architecture for the Balance System system. This diagram shows the major external interfaces that provide the capabilities of Balance System.

The Balance System is a single player game. The game is a standalone game played on the STM32 board. This system would be a game where the user would have to balance a ball on a LCD screen that is builtin on the STM32 board. The objective of the game is to balance the ball on the screen based on the way the board was tilted. Using an onboard gyroscope that communicates across the board to the MCU though SPI communication. Balance System would keep track of the current position of the ball and where the next updated move is. This helps keep track of the system of where the ball is until a movement has occurred. Balance System shall process at a maximum 180 Hz. This would give the user enough time to process the current angle of the ball and be able to present on the LCD-TFT screen.

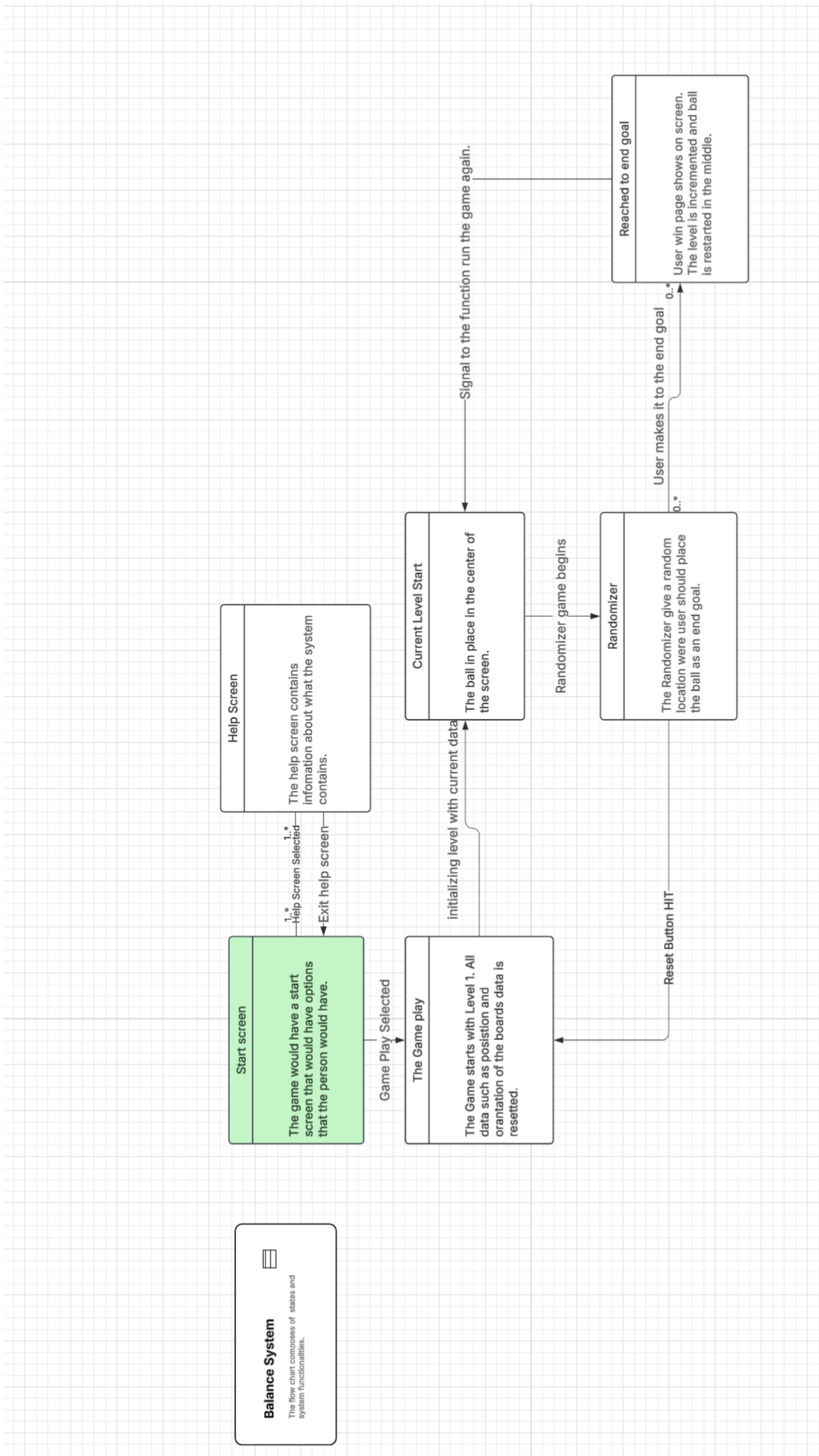

Figure 1: System Overview

1.3 Document Overview

This section provides information about this document's security/privacy considerations, contents, structure, and version information. This section also provides information regarding how specifications are formatted in this artifact and how they can best be understood.

Because this is the overall system performance specification, this document may provide traceability to miscellaneous project documents. This allows for tracking of related doctrine, vendor, and draft specification requirements as the document is being created.

CHAPTER 2

References

This section provides a list of referenced items for this document.

2.1 Acronyms and Abbreviations

This section defines acronyms and abbreviations used in this and related documents.

Table 1: Acronym Definitions

Acronym	Definition
.AUX	LaTeX Auxiliary Files
.CSV	Excel Comma Separated Variable
.PDF	Portable Document Format
.PS	Portable Document Format
ABIS	Automated Biometric Identification System
BioAPI	BioAPI
DIDs	Data Item Descriptions
DOORS™	DOORS
Gantt	Gantt Chart
Garmin	Garmin Inc.
HMI	Human-Machine Interface
JIDA	Joint Improvised-threat Detection Agency
KPP	Key Performance Parameter
KSA	Key System Attribute
MISO	Military Information Support Operations
OCDs	Operational Concept Descriptions
SOO	Statement Of Objectives
SOW	Statement Of Work
SPSs	System Performance Specifications
SRD	System Requirements Document
SRSs	System Requirements Specifications
SSSs	System / Subsystem Specifications

Acronym definitions continue on next page

Acronym definitions – continued from previous page	
Acronym	**Definition**
STS	System Test Specification
TOC-L	Tactical Operations Center - Light
UAV	Uncrewed Aerial Vehicle
USAF	United States Air Force
USArmy	United States Army
USMC	United States Marine Corp
USN	United States Navy
USSF	United States Space Force
End of acronym definition table	

2.2 Glossary and Definitions

This section defines glossary terms used in this and related documents.

Table 2: Glossary Terms and Definitions

Glossary Term	Definition
STM32F429I	Micro-controller board has all component fit onto one board.
Customer	The professor that is view the grading all assignments.
End of glossary terms table	

2.3 Referenced Documents

This section lists the referenced documents for this document. The references are categorized into two categories:

External Documents not directly associated with this project.

Project Documents that are directly associated with this project.

2.3.1 External Documents

2.3.2 Project Specific Documents

CHAPTER 3

Requirements

This section provides the requirements that drive the design and implementation of the Balance System. These specifications are divided into the major segments of the system. Each requirement is listed in the segment that provides the specified capability, thus this section provides an immediate mapping of implementation requirements to the segment in which each requirement is met. Each requirement also includes traceability to high-level requirements that drive the specific capability. Validation methodology is provided here but verification traceability is provided in the **STS** artifacts.

The requirements also are specified in an order that generally allows for all precursor requirements to be stated before they are needed by a successor requirement. Thus, States and Modes are defined at the onset so that they can be used to regulate when external interfaces and processing steps may occur. Likewise, external interfaces are described so that the data from the interfaces may be used in, or created by, the ensuing processing. Once the processing is specified, the internal interface and data requirements are listed, showing how the overall system segments tie together. The remainder of the sections follow a somewhat similar pattern, but these latter sections contain disparate requirements that are separated and organized in a standard way so the contents can be easily scanned to locate specific requirements based on their type and expected location within the **SSS**.

These specifications also include qualifications for both Threshold (must meet) and Objective (want to meet) requirements for Balance System. The reader is cautioned to ensure that the requirement details be understood for the two modifiers.

3.1 States and Modes

This section lists the states, sub-states, modes, and sub-modes that are provided by the system. While these terms can be construed in many ways, for this document, the following meanings are used:

States are the basic configurations of the system.

Sub-states are the effective state of being for the system.

3.1.1 States

A summary of the states is provided in Table 3. See the formal specifications, if applicable, in the following sections for formal statement of the state requirements, and accompanying notes that provide further clarification on the meanings of the states.

STATES	
State Name	**Summary**
State 1	Main Page 3111
State 2	Help Page 3112
State 3	Main Game Mode 3121
State 4	Balls movement 3131
State 5	Ball matches target 3132
State 6	Displaying level 3133
State 7	Displaying time 3134
State 8	Changing levels 3141
State 9	Failed level 3142
State 10	Hard turns 3143

Table 3: Summary of States for Balance System

Specification 3.1.1.1 State One	
(KPP) Text	The Balance System will have an opening screen that the game should also start with. This will allow the user to select viewing details of how to play the game or just to play the game.
Status	**Phase 1** Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	**SPS** This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.

Specification 3.1.1.2 State Two	
(KSA) Text	The Balance System shall have a separate page that give clear indication on how the game shall be played. This page shall take the whole lCD screen to display.
Status	**Phase 1** Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	**SPS** This requirement is referenced in the SPS sheet.
Notes	N/A This requirement is referenced in the SPS sheet.

Specification 3.1.1.3 State Three	
Text	The Balance System shall have a consistent location where the ball within each level will have the ball placed. This place will be noticeable and be accessible for any player to move the ball in the correct location..
Status	**Phase 1** Levels must transition to the next live after every win.
Acceptance	This requirement shall be verified by demonstration.
Traceability	**SPS** This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.

Specification 3.1.1.4 State Four	
Text	”The Balance System shall have a transitional model that would change the position on the ball based off the MEMs reading. The reading from the Mem control should describe the movement of the ball”
Status	**Phase 1** Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	**SPS** This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.

Specification 3.1.1.5 State Five	
Text	The Balance System shall a win page once the user matched the write location for where the ball should be landing.
Status	**Phase 1** Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	**SPS** This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.

Specification 3.1.1.6 State Six	
Text	The Balance System shall have a level score while the game is running. This would be a running counter that is displayed once the user starts the game.
Status	**Phase 1** Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	**SPS** This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.

Specification 3.1.1.7 State Seven	
Text	The Balance System shall provide a timer on the top left of the game to keep track of level to keep track of time per level. This is an ideal location for the user see there time limit.
Status	**Phase 1** Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	**SPS** This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.

Specification 3.1.1.8 State Eight	
Text	The Balance System shall change levels once the user reached a certain level.
Status	**Phase 1** Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	**SPS** This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.

Specification 3.1.1.9 State Three	
Text	The Balance System shall go back main menu once the user does not make the mark of the user.
Status	**Phase 1** Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	**SPS** This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.

Specification 3.1.1.10 State Ten	
Text	The Balance System shall discard sharp changes while the microcontroller is being tilted.
Status	**Phase 1** Currently being worked on.
Acceptance	This requirement shall be verified by demonstration.
Traceability	**SPS** This requirement is referenced in the SPS sheet.
Notes	1. The State-3 state generalizes the case where the system is working with the requirement.

3.2 External Interfaces

The external interfaces for this system are shown in Figure 2. The requirements for these interfaces are described in more detail in the following sections.

Network The network(s) that connect to the Balance System, § 3.2.1

Power The network(s) that connect to the Balance System, § 3.2.1

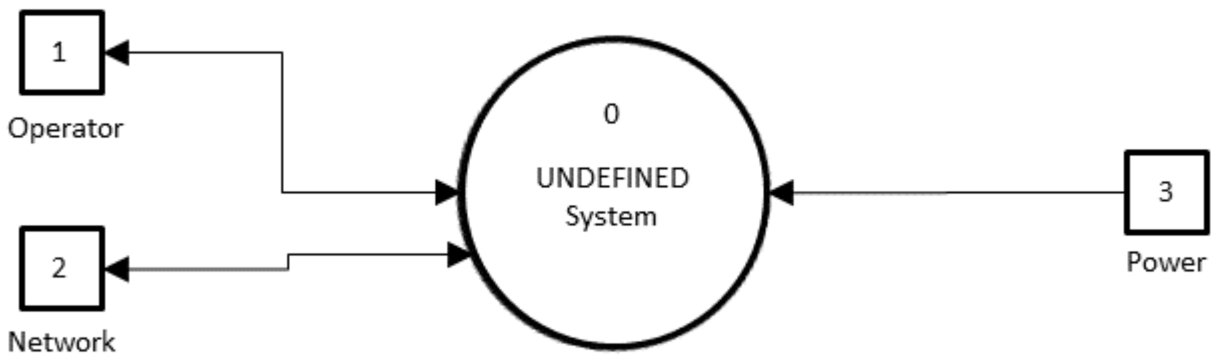

Figure 2: System Context Diagram (DFD-C)

3.2.1 Network Interfaces

Specification 3.2.1.1 Approved Network	
(KPP) Text	The current network would not be needed for completion of the system.
Status	**Phase 1** The system must work independent from the main system.
Acceptance	The system would not need to be dependency from any network communication.
Traceability	**N/A** The base requirement comes from the STP document.
Notes	1. N/A

3.2.2 Power Interfaces

Specification 3.2.2.1 Power	
(KPP) Text	All Balance System variants shall be capable of connecting to **...TBD...** Power.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This requirement is a base requirement.
Notes	1. N/A

3.3 Capabilities

This section defines the capability areas for the Balance System. The segment design is structured to meet the requirements as specified in the **...TBD...** artifacts. Each area provides a subset of the overall capabilities for the Balance System segments. These segments are shown in Figure 3, are summarized below, and are more fully specified in the following subsections.

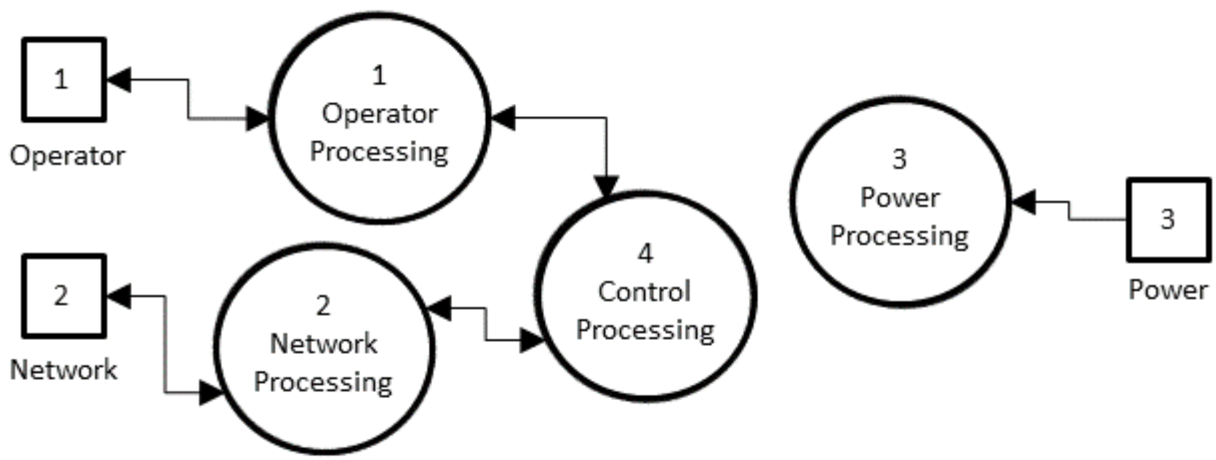

Figure 3: System Top-Level Diagram (DFD-0)

The capability requirements for these segments are described in more detail in the following sections:

Operator Processing handles the **HMI** interface to the operator and provides overall control and configuration to the Balance System, § 3.3.1.

Network Processing handles the network interface, § 3.3.2.

Power Processing handles the power input and conversions as necessary, § 3.3.3.

Control Processing handles all major capability control, § 3.3.4.

3.3.1 Operator Processing

The operator requirements for Balance System are listed below.

Specification 3.3.1.1 Power	
(KPP) Text	All Balance System variants shall be capable of **...TBD...** operator inputs.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files, e.g., OperatorInputs.tex and OperatorOutputs.tex, as needed. Just use the RequirementNumberAM and RqtNumberBase commands to keep numbers correct if subsubsections are added.

Specification 3.3.1.2 Power	
(KPP) Text	All Balance System variants shall be capable of **...TBD...** operator outputs.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files, e.g., OperatorInputs.tex and OperatorOutputs.tex, as needed. Just use the RequirementNumberAM command to keep numbers correct if subsubsections are added.

3.3.2 Network Processing

The network requirements for Balance System are listed below.

Specification 3.3.2.1 Network Types	
(KPP) Text	All Balance System variants shall be capable of **...TBD...** network types.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files/folders, e.g., NetworkTypes.tex, NetworkInputs.tex, and NetworkOutputs.tex, etc. as needed. Just use the RequirementNumberAM and RqtNumberBase commands to keep numbers correct if subsubsections are added.

Specification 3.3.2.2 Network Inputs	
(KPP) Text	All Balance System variants shall be capable of **...TBD...** network inputs.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files/folders, e.g., NetworkTypes.tex, NetworkInputs.tex, and NetworkOutputs.tex, etc. as needed. Just use the RequirementNumberAM and RqtNumberBase commands to keep numbers correct if subsubsections are added.

Specification 3.3.2.3 Network Outputs	
(KPP) Text	All Balance System variants shall be capable of **...TBD...** network outputs.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files/folders, e.g., NetworkTypes.tex, NetworkInputs.tex, and NetworkOutputs.tex, etc. as needed. Just use the RequirementNumberAM and RqtNumberBase commands to keep numbers correct if subsubsections are added.

3.3.3 Power Processing

The power requirements for Balance System are listed below.

Specification 3.3.3.1 Power Voltage	
(KPP) Text	All Balance System variants shall be capable of **...TBD...** power voltage(s).
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files/folders as needed. Just use the RequirementNumberAM and RqtNumberBase commands to keep numbers correct if subsubsections are added.

Specification 3.3.3.2 Power Current	
(KPP) Text	All Balance System variants shall be capable of **...TBD...** power current.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files/folders as needed. Just use the RequirementNumberAM and RqtNumberBase commands to keep numbers correct if subsubsections are added.

3.3.4 Control Processing

The control requirements for Balance System are listed below.

Specification 3.3.4.1 Control One	
(KPP) Text	All Balance System variants shall be capable of **...TBD...** control one.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files/folders as needed for areas of control capabilities. Just use the RequirementNumberAM and RqtNumberBase commands to keep numbers correct if subsubsections are added.

Specification 3.3.4.2 Control Two	
(KPP) Text	All Balance System variants shall be capable of **...TBD...** control two.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This requirement is a base requirement.
Notes	1. Add as many of these as necessary. Split into files/folders as needed for areas of control capabilities. Just use the RequirementNumberAM and RqtNumberBase commands to keep numbers correct if subsubsections are added.

3.4 Internal Interface Requirements

This section provides the internal interface requirements. These requirements for these interfaces are described in more detail in the following sections:

Internal Interface Requirement One stuff

Internal Interface Requirement One more stuff

3.5 Internal Data Requirements

This section provides the internal data requirements. The Balance System capability is segmented into the following specification groups:

Data Storage provides the data storage requirements, § 3.5.1.

Report Logs provides the report log requirement, § 3.5.2.

3.5.1 Data Storage

Specification 3.5.1.1 Data Storage	
(KSA) Text	All Balance System variants shall store digital files received for transmission using 2 TB of internal storage.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This is a base requirement.
Notes	1. N/A

Specification 3.5.1.2 Information Transport	
(KSA) Text	All Balance System variants shall be able to manually upload and download digital files using external SD card, CD, and DVD formats.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This is a base requirement.
Notes	1. N/A

3.5.2 Report Logs

Specification 3.5.2.1 Report Logs	
(KSA) Text	The system shall locally store reports for up to 12 months and be capable of exporting in .txt, .csv, and .xml formats.
Status	**Phase 1** T=O
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This is a base requirement.
Notes	1. Where does this information come from? **...TBD...**.

3.6 Adaptation Requirements

Later on the next idea is add legs to the system.

3.7 Safety Requirements

This section lists the safety requirements for the system. The Balance System capability is segmented into the following specification groups:

Electromagnetic Radiation describes the safety requirements pertaining to the presence of EMR, § **??**.

3.8 Security and Privacy Requirements

This section provides the security and privacy requirements for Balance System. The Balance System capability is segmented into the following specification groups:

Security Requirements provides the physical and cyber security requirements of the system, § 3.8.1.

Privacy Requirements provides the privacy requirements of the system, § 3.8.2.

3.8.1 Security Requirements

3.8.1.1 Physical Security

Specification 3.8.1.1.1 Anti-Tamper	
Text	The Balance System shall deter all unauthorized alterations, countermeasure development, and system exploitation.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by testing.
Traceability	**N/A** This is a base requirement.
Notes	1. N/A

3.8.1.2 Cyber Security

This section is provided for future expansion.

3.8.2 Privacy Requirements

This section is provided for future expansion.

Specification 3.8.1 Privacy	
Text	The system shall **...TBD...**.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This is a base requirement.
Notes	1. **...TBD...**.

3.9 Environmental Requirements

This section defines the environmental requirements for Balance System.

3.10 Technology Resource Requirements

This section provides the overall technology resource requirements for the system. These capabilities are divided into the following sections:

Hardware details about the hardware to be used.

Software details about the software to be used.

Communications details about the communications to be used.

Other details about other technology resource requirements not covered above.

Utilization details about the resource utilization.

3.11 System Quality Requirements

This section specifies the Balance System quality requirements.

3.11.1 Quality Systems

Specification 3.11.1.1 Development Quality	
Text	The system design and development shall follow the implementers' certified processes.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by inspection.
Traceability	**N/A** This requirement is a base requirement.

3.11.2 Operational Quality

Ensure that the system would load properly connects with the correct subsystem.

3.11.3 Quantitative Metrics

3.11.3.1 Object Detection Metrics

3.11.3.2 Object Identification Metrics

3.11.4 Qualitative Metrics

3.11.4.1 Media Selection Metrics

This Section is **...TBD...**.

3.12 Design and Construction Requirements

This section provides the Balance System design and construction requirements.

3.12.1 Regulatory Restrictions

This section is included for future expansion.

Specification 3.12.1.2 Proprietary Components	
Text	All Balance System variants shall include only software components that are open source or to which the developer has unlimited rights.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by inspection.
Traceability	**N/A** This requirement is a base requirement.
Notes	1. The inspection is of the software design documents and build scripts to ensure that all code meets this requirement.

3.12.2 Design Defences

This section provides the construction constraints.

This section is provided for future expansion.

3.12.3 Construction Constraints

This section provides the construction constraints.

This section is provided for future expansion.

3.13 Personnel Requirements

This section is provided for future expansion.

3.14 Training Requirements

3.14.1 Manuals

Specification 3.14.1.1 Operator's Guide	
Text	The Balance System training shall provide an operator's manual.
Status	**Phase 1** T=O
Acceptance	This requirement shall be verified by inspection.
Traceability	**N/A** This is a base requirement.
Notes	1. N/A

3.14.2 Materials

Specification 3.14.2.1 Training Materials	
Text	The Balance System training shall provide all training course materials.
Status	**Phase 1** T=O
Acceptance	This requirement shall be verified by inspection.
Traceability	**N/A** This is a base requriement.
Notes	1. N/A

3.14.3 Courses

Specification 3.14.3.1 Training Courses	
Text	The vendor will provide specific training and course materials and programs of instruction.
Status	**Phase 1** T=O
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This is a base requirement.
Notes	1. N/A

3.15 Logistics Requirements

3.15.1 Transportability

Specification 3.15.1.1 Transportability	
(KSA) Text	All Balance System variants shall be transportable by air, ground, and maritime resources.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This is a base requirement.
Notes	1. N/A

3.16 Packaging Requirements

3.16.1 Shipping Constraints

This section is provided for future expansion.

3.17 Other Requirements

3.17.1 Broadcast Playlist Manager

Specification 3.17.1.1 Playlist Manager	
(KSA) Text	All Balance System variants shall interface with an approved external broadcast playlist manager that will provide the operator the ability to manage programming in real-time.
Status	**Phase 1** T=O
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This is a base requirement.
Notes	1. N/A

3.17.2 Information Exchange

Specification 3.17.2.1 IP Data	
(KPP) Text	All Balance System variants shall be capable of disseminating data on an approved network with a reasonable response time of less than four (4) hours.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by demonstration.
Traceability	**N/A** This is a base requirement.
Notes	1. N/A

3.18 Precedence of Requirements

3.18.1 Safety

Specification 3.18.1.1 Safety Requirements Precedence	
Text	All Balance System variants shall meet safety requirements listed in Section 3.7 before all other requirements.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by inspection.
Traceability	**N/A** This requirement is a base requirement.
Notes	1. Obviously safety is of utmost importance. 2. The inspection is of design notes and rationale whereby design decisions relating to precedence are recorded.

3.18.2 Security and Privacy

Specification 3.18.2.1 Security Requirements Precedence	
Text	All Balance System variants shall meet security requirements listed in Section 3.8.1 before all other requirements with the exception of safety.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by inspection.
Traceability	**N/A** This requirement is a base requirement.
Notes	1. Security trumps privacy since good security should help ensure privacy. 2. The inspection is of design notes and rationale whereby design decisions relating to precedence are recorded.

Specification 3.18.2.2 Privacy Requirements Precedence	
Text	All Balance System variants shall meet privacy requirements listed in Section 3.8.2 before all other requirements with the exception of safety and security.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by inspection.
Traceability	**N/A** This requirement is a base requirement.
Notes	1. Privacy is trumped by security since good security should help ensure privacy. 2. The inspection is of design notes and rationale whereby design decisions relating to precedence are recorded.

3.18.3 Other

Specification 3.18.3.1 Other Requirements Precedence	
Text	All Balance System variants shall meet with equal precedence all other requirements not pertaining to safety, security, and privacy.
Status	**Phase 1** Threshold
Acceptance	This requirement shall be verified by inspection.
Traceability	**N/A** This requirement is a base requirement.
Notes	1. The inspection is of design notes and rationale whereby design decisions relating to precedence are recorded.

CHAPTER 4

Qualification Provisions

The qualification provisions are listed in the acceptance row of the specifications in Section 3.

CHAPTER 5

Traceability

This section provides a list of the sources, if applicable, for each requirement. This traceability connects the specifications in this document to those presented in higher level sources such as a Joint Urgent Operational Need (**JUON**) document, Joint Emergent Operational Need (**JEON**), or a STATEMENT OF WORK (**SOW**)

The traceability of all specifications from each requirement to its source, if applicable, is listed in the specifications presented in section 3. Traceability from each document to requirements is provided below.

Table 4: Source to Requirement Traceability.

Source Requirement	**Traced Requirement**
...TBD...:: 4.1.2	**??**
N/A :: This is a base requirement.	3.5.1.1
N/A :: This is a base requirement.	3.5.1.2
N/A :: This is a base requirement.	3.5.2.1
N/A :: This is a base requirement.	**??**
N/A :: This is a base requirement.	**??**
N/A :: This is a base requirement.	3.8.1.1.1
N/A :: This is a base requirement.	3.8.1
N/A :: This is a base requirement.	3.14.1.1
N/A :: This is a base requirement.	3.14.3.1
N/A :: This is a base requirement.	3.15.1.1
N/A :: This is a base requirement.	3.17.1.1
N/A :: This is a base requirement.	3.17.2.1
N/A :: This is a base requriement.	3.14.2.1
N/A :: This requirement is a base requirement.	**??**
N/A :: This requirement is a base requirement.	**??**

Continued on next page

Table 4 – continued from previous page

Source Requirement	**Traced Requirement**
N/A :: This requirement is a base requirement.	**??**
N/A :: This requirement is a base requirement.	**??**
N/A :: This requirement is a base requirement.	**??**
N/A :: This requirement is a base requirement.	**??**
N/A :: This requirement is a base requirement.	**??**
N/A :: This requirement is a base requirement.	3.2.1.1
N/A :: This requirement is a base requirement.	3.2.2.1
N/A :: This requirement is a base requirement.	3.3.1.1
N/A :: This requirement is a base requirement.	3.3.1.2
N/A :: This requirement is a base requirement.	3.3.2.1
N/A :: This requirement is a base requirement.	3.3.2.2
N/A :: This requirement is a base requirement.	3.3.2.3
N/A :: This requirement is a base requirement.	3.3.3.1
N/A :: This requirement is a base requirement.	3.3.3.2
N/A :: This requirement is a base requirement.	3.3.4.1
N/A :: This requirement is a base requirement.	3.3.4.2
N/A :: This requirement is a base requirement.	3.12.1.2
N/A :: This requirement is a base requirement.	3.18.1.1
N/A :: This requirement is a base requirement.	3.18.2.1
N/A :: This requirement is a base requirement.	3.18.2.2
N/A :: This requirement is a base requirement.	3.18.3.1
KPP-1 :: [**ref˙KNEAD˙Manual**]	3.1.1.1
KPP-1 :: [**ref˙KNEAD˙Manual**]	**??**
KSA-1 :: [**ref˙KNEAD˙Manual**]	3.1.1.2
KSA-2 :: [**ref˙KNEAD˙Manual**]	**??**
Multi-3 :: [**ref˙KNEAD˙Manual**]	**??**
Single-3 :: [**ref˙KNEAD˙Manual**]	3.1.1.9

APPENDIX A

Notes

This section provides notes, as necessary, to document the system segmentation specification.

APPENDIX B

Key Performance Parameters and System Attributes

This Appendix provides the key performance parameters and key system attributes, summarized in a short list for easy review.

B.1 Key Performance Parameters

Table B.1: Key Performance Parameter Specifications

Specification	**Key Performance Parameter**
3.1.1.1	
??	Thingie one. Thingie two. Thingie three.
??	
3.2.1.1	
3.2.2.1	
3.3.1.1	
3.3.1.2	
3.3.2.1	
3.3.2.2	
3.3.2.3	
3.3.3.1	
3.3.3.2	
3.3.4.1	
3.3.4.2	
3.17.2.1	

B.2 Key System Attributes

Table B.2: Key System Attribute Specifications

Specification	Key System Attribute
3.1.1.2	
??	Thingamabobs one. Thingamabobs two. Thingamabobs three.
3.5.1.1	
3.5.1.2	
3.5.2.1	
3.15.1.1	
3.17.1.1	

Index